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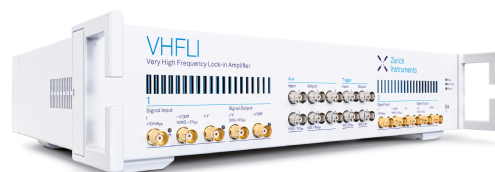
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Cite as: J. Appl. Phys. **135**, 193105 (2024); doi: [10.1063/5.0174179](https://doi.org/10.1063/5.0174179)

Submitted: 13 February 2024 · Accepted: 3 May 2024 ·

Published Online: 15 May 2024



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ABSTRACT

Asymmetric quantum wells (AQWs) utilizing interband transitions enhance second-order susceptibility over a wide wavelength range compared to natural crystals. The nonlinear susceptibility is further enhanced in AQWs with type-II band alignment as compared to type-I band alignment, a result of the larger interband charge shift. This enhancement is demonstrated in this work by analyzing three type-I and type-II AQW designs based on the lattice-matched InP/AlGaInAs materials systems using the envelope wavefunction approximation. The calculated interband second-order susceptibility tensor elements in type-II structures range between 20 and 1.60×10^3 pm/V for nearly resonant optical rectification and difference frequency generation applications at near-infrared and terahertz wavelengths, an improvement of nearly 1 order of magnitude over the type-I structures and 1–2 orders of magnitude over natural crystals such as LiNbO₃, KTiOPO₄ (KTP), or GaAs. A factor of 2–3 further enhancement of the tensor elements is achieved by optimizing the well widths and band offsets of the type-II asymmetric quantum wells. The type-II structure can be implemented in other material systems spanning the longwave infrared to visible wavelengths, enhancing nonlinear susceptibility for various applications, including photonic integrated circuits.

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I. INTRODUCTION

Conceived in the 1960s, material development for nonlinear optics has a long history and great potential. The second-order nonlinear susceptibility $\chi^{(2)}$ is the lowest order and hence the strongest among optical nonlinearities and is used extensively in frequency up/down conversion, parametric generation,¹ and electro-optic modulation.² While virtually all materials, crystalline or amorphous, possess the third-order nonlinear susceptibility $\chi^{(3)}$, only crystalline materials without centro-symmetry have substantial $\chi^{(2)}$ such as ZnSe, GaAs, and InP.³ Unfortunately, due to the difficulty of achieving phase matching (PM), these crystals are not widely used. The existing bulk phase-matchable nonlinear crystals such as LiNbO₃ or KTiOPO₄ have $\chi^{(2)}$ values far from theoretical upper bounds,^{4,5} especially in the near-IR (NIR) to ultraviolet (UV) range. Artificial structures⁶ such as asymmetrical quantum wells (AQWs) with inter-subband (ISB) transitions^{7–9} have been proposed as an effective way

to achieve high $\chi^{(2)}$ by engineering the asymmetric charge distributions, emulating polar bonds on a larger length scale.^{8–13} However, current $\chi^{(2)}$ enhancement achieved in III-V AQWs^{14,15} is limited to the mid-IR (MIR) due to small band offsets,^{13,14,16} and the palette of functional materials with large $\chi^{(2)}$, especially in the NIR to UV range remains sparse since the 1990s.

ISB $\chi^{(2)}$ nonlinearities require high doping,^{16,17} are prone to large absorption near resonance,^{8,18} and only engender one component of $\chi^{(2)}$ tensor, $\chi_{zzz}^{(2)} = d_{33}$.^{1,7} For this reason, as early as 1987 it was proposed to extend the concept of envelope function engineering to the interband (IB) transition in AQWs.^{19,20} The interband nonlinearity expands the operational frequency range all the way to the bandgap energy; however, understanding the design and optimization of the $\chi^{(2)}$ requires extra effort because of the cancellation effects, as virtual transitions by electrons or holes tend to cancel each other. This is because the distance between the centers of charge of the lowest CB and highest VB subbands $\Delta z_{11} = (\int |\Psi_{c,1}|^2 - \int |\Psi_{v,1}|^2) z dz$

(interband charge shift) is very small.^{1,20,21} Therefore, using type-I asymmetric stepped QWs⁷ (T1 ASQW) or asymmetric coupled quantum wells^{16,20} (ACQW) previously studied for ISB $\chi^{(2)}$ does not result in large interband $\chi^{(2)}$.

This work proposes the use of asymmetric quantum wells with type-II band alignment (T2 AQWs) to increase $\chi^{(2)}$, establishes a theoretical framework for envelope wavefunction engineering of T2 AQW designs, and compares the T2 structures with a T1 ASQW and ACQW to quantify the enhancement in $\chi^{(2)}$ resulted from the differing band alignments. The T2 AQWs enhance the strength of the interband nonlinearity by breaking the symmetry between the conduction and valence band edge potential profiles. In a T2 AQW, the electrons in the lowest energy subband and holes in the highest energy subband are confined in separate well layers, increasing the interband charge shift Δz_{11} and the electric dipole moment generated when the electron makes a transition (real or virtual) between the bands.

The lattice-matched InP/AlInAs/AlGaInAs material system is used to demonstrate enhanced $\chi^{(2)}$ for nearly resonant optical rectification in the NIR and difference frequency generation in the terahertz (THz). The results are also broadly applicable to other semiconducting material systems with proper type-II offsets between conduction and valence bands and to other second-order nonlinear processes such as second harmonic generation or sum frequency generation.

II. $\chi^{(2)}$ BASED ON MIXED INTERBAND AND INTERSUBBAND TRANSITIONS IN QUANTUM WELLS

A schematic of T1 and T2 AQWs made of a model materials system is shown in Fig. 1. The two well regions in the T2 AQW form (i) an electron well with thickness l_1 and (ii) a hole well with thickness l_2 . We can consider one individual QW by using semi-infinite thick barriers approximation to neglect the interwell coupling. The conduction and valence band edges $E_{c1}, E_{c2}, E_{v1}, E_{v2}$ and E_{cb}, E_{vb} are defined for the electron well (well 1), hole well (well 2) and barriers, correspondingly. For simplicity, keep all the band offsets between the barriers and well layers constant and use six parameters to specify the AQW: well widths l_1 and l_2 for well 1 and well 2, and conduction and valence band edge offset $\Delta E_{c2,c1}, \Delta E_{v2,v1}, \Delta E_{c1,cb}$, and $\Delta E_{v1,vb}$ between the two wells and the barriers. The key property of the T2 AQW is the type-II alignment between the well layers. The type-II alignment breaks the symmetry between conduction and valence band edges and results in spatial separation of the electron (e_1) and heavy-hole/light-hole (hh_1/lh_1) envelope wavefunction peaks. Schematically, this is shown by the charge shift $\Delta z_{11} = (\int |\Psi_{c1}|^2 - \int |\Psi_{v1}|^2) dz$.

The envelope wavefunctions for the quantum-confined electrons and holes are denoted as $\psi_{b,g}(z)$ with band index $i = 1, 2, 3$ for heavy holes (hh), electrons (e), and light holes (lh) respectively, and the subband index $g = 1, 2, 3 \dots$. The envelope wavefunctions in the AQW are determined by solving the one-dimensional time-independent Schrödinger equation using the finite difference method.²² MATLAB R2020b is used to calculate the eigenfunctions and eigenenergies of the diagonal Hamiltonian matrix.

The second-order nonlinear susceptibility tensor element $\chi_{ijk}^{(2)}$ of the quantum well is given as¹⁹

$$\chi_{ijk}^{(2)}(\omega_1 + \omega_2, \omega_1, \omega_2) = \frac{N_z}{2\epsilon_0 \hbar^2 L^2} \mathcal{P}_I \sum_{b_1, b_2, b_3} \sum_{g, m, n} f_{b_1 g}(\mathbf{k}) \times \sum_{k_i} \frac{\mu_{12, gm}^i \mu_{23, mn}^j \mu_{31, ng}^k}{(\omega_{21, mg}(\mathbf{k}_i) - \omega_1 - \omega_2)(\omega_{31, ng}(\mathbf{k}_i) - \omega_2)}. \quad (1)$$

In this formula, the dipole moment $\mu_{12, gm}$ represents each transition doublet between bands b_1, b_2 and subbands g, m and the triple products of dipole moments on the numerator are calculated approximately using the relation between the square of momentum matrix elements $|\hat{x} \times \mathbf{p}_{12, gm}|^2$ or $|\hat{z} \times \mathbf{p}_{12, gm}|^2$.²³ N_z is the number

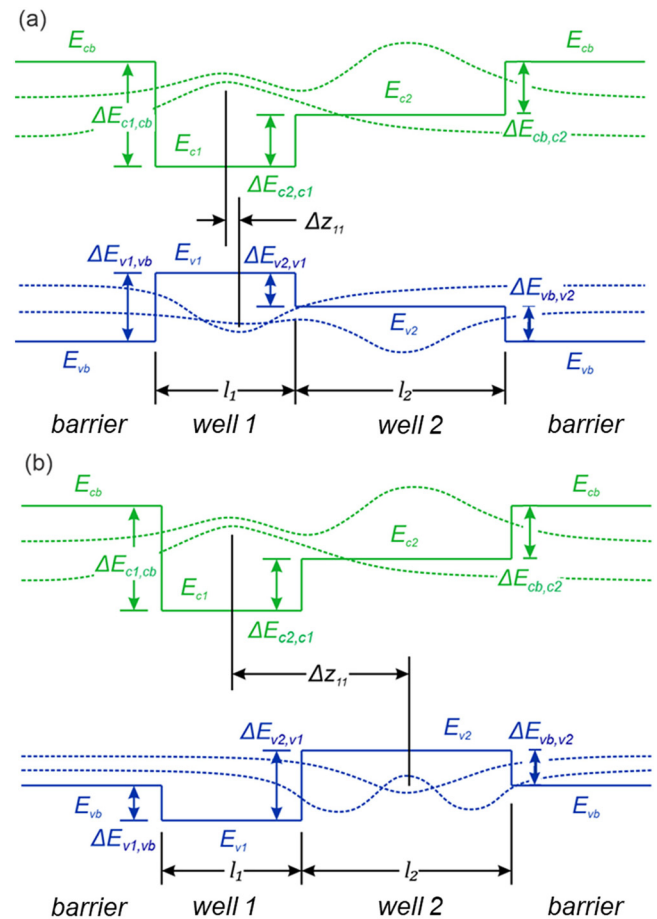


FIG. 1. (a) Type-I asymmetric quantum well (T1 AQW) and (b) type-II asymmetric quantum well (T2 AQW) schematic structures. Conduction (valence) band offsets are shown as green (blue) lines. Electron well is denoted by layer 1 with width l_1 and hole well is denoted by layer 2 with width l_2 . The band offsets of both barriers are identical, with semi-infinite thickness.

of quantum well periods per unit length. $f_{b_1,g}(\mathbf{k})$ is the Fermi function for band b_1 and subband g . The complex angular frequency $\omega_{21,mg}(\mathbf{k})$ for transitions from b_2, m to b_1, g is given by the difference between bound state energies,

$$\omega_{21,mg}(k_t) = \frac{E_{b_2,m}(k_t) - E_{b_1,g}(k_t)}{\hbar} - i\frac{\Gamma_{b_2,m}}{2}, \quad (2)$$

where $\Gamma_{b_2,m}$ is a dipole dephasing rate that is used to model damping phenomena as a product of various scattering and

recombination mechanisms.^{24,25} The intrinsic permutation operator $\mathcal{P}_I$ sums over all permutations of $(-\omega_1 - \omega_2, i)$, (ω_1, j) , and (ω_2, k) . Materials that cannot be adequately described within this formalism, e.g., those exhibiting a large concentration of mid-gap defect states, are not suitable candidates for engineering large $\chi^{(2)}$ within the envelope wavefunction formalism.

Expressing Eq. (1) in integral form in terms of the isotropic in-plane wavevector k_t yields

$$\chi_{ijk}^{(2)}(\omega_1 + \omega_2, \omega_1, \omega_2) = \frac{N_z}{\pi\epsilon_0\hbar^2} \mathcal{P}_I \sum_{b_1,b_2,b_3} \sum_{g,m,n} \int f_{b_1,g}(k_t) \frac{\mu_{12,gm}^i \mu_{23,mn}^j \mu_{31,ng}^k}{(\omega_{21,mg}(k_t) - \omega_1 - \omega_2)(\omega_{31,ng}(k_t) - \omega_2)} k_t dk_t. \quad (3)$$

For intrinsic material with a Fermi level in the middle of the band gap, the Fermi function $f_{b_1,g}(k_t)$ is unity for valence bands $b_1 = hh$ or lh and is zero for conduction band $b_1 = e$. Equation (3) can be further simplified as

$$\begin{aligned} \chi_{ijk}^{(2)}(\omega_1 + \omega_2, \omega_1, \omega_2) &= \frac{N_z}{\pi\epsilon_0\hbar^2} \mathcal{P}_I \sum_{hh,b_2,b_3} \sum_{g,m,n} \int \frac{\mu_{12,gm}^i \mu_{23,mn}^j \mu_{31,ng}^k}{(\omega_{21,mg}(k_t) - \omega_1 - \omega_2)(\omega_{31,ng}(k_t) - \omega_2)} k_t dk_t \\ &+ \frac{N_z}{\pi\epsilon_0\hbar^2} \mathcal{P}_I \sum_{lh,b_2,b_3} \sum_{g,m,n} \int \frac{\mu_{12,gm}^i \mu_{23,mn}^j \mu_{31,ng}^k}{(\omega_{21,mg}(k_t) - \omega_1 - \omega_2)(\omega_{31,ng}(k_t) - \omega_2)} k_t dk_t. \end{aligned} \quad (4)$$

Near the electron-heavy hole interband resonance, the summation in Eq. (4) is dominated by the product of the interband dipole matrix elements and the charge shift Δz_{11} between the first electron and heavy hole envelope wavefunctions, with

$$\begin{aligned} \chi_{ijk}^{(2)}(\omega_1 + \omega_2, \omega_1, \omega_2) &\approx \frac{N_z}{3\pi\epsilon_0\hbar^2} \langle u_{c,1} | x | u_{hh,1} \rangle^2 \int \frac{\langle \psi_{e,1} | \psi_{hh,1} \rangle \langle \psi_{hh,1} | \psi_{e,1} \rangle [\langle \psi_{e,1} | z | \psi_{e,1} \rangle - \langle \psi_{hh,1} | z | \psi_{hh,1} \rangle]}{(\omega_{21,mg}(k_t) - \omega_1 - \omega_2)(\omega_{31,ng}(k_t) - \omega_2)} k_t dk_t \\ &= \frac{N_z}{3\pi\epsilon_0\hbar^2} \langle u_{c,1} | x | u_{hh,1} \rangle^2 \int \frac{\langle \psi_{e,1} | \psi_{hh,1} \rangle \langle \psi_{hh,1} | \psi_{e,1} \rangle \Delta z_{11}}{(\omega_{21,mg}(k_t) - \omega_1 - \omega_2)(\omega_{31,ng}(k_t) - \omega_2)} k_t dk_t. \end{aligned} \quad (5)$$

$\chi^{(2)}$ is strongly dependent on charge shift, clearly showing the advantages of T2 AQW structure. This structure is advantageous for nearly resonant optical rectification (OR) and difference frequency generation (DFG) applications in the far IR and THz spectrum where powerful laser sources are not commonly available.^{26,27} Details of derivation can be found in the [supplementary material](#).

III. DISCUSSION

Three types of asymmetric quantum well structures were compared: (i) asymmetric coupled quantum well (ACQW), (ii) type-I asymmetric stepped quantum well (T1 ASQW), and (iii) type-II asymmetric quantum well (T2 AQW). These structures' $\chi^{(2)}$ values depend on electric dipole moments, envelope wavefunction overlaps, and transition frequencies $\omega_{21,mg}(\mathbf{k})$. To distinguish differences solely due to the quantum well potential profile, a two-step design process was employed. Initially, lattice-matched InP/Al_{0.38}Ga_{0.10}In_{0.52}As T2 AQW well widths were optimized for maximum $\chi^{(2)}$ in nearly resonant optical rectification. Then, well widths were selected for lattice-matched Al_{0.35}Ga_{0.12}In_{0.53}As/Al_{0.48}In_{0.52}As ACQW and

Al_{0.27}Ga_{0.20}In_{0.53}As/Al_{0.42}Ga_{0.06}In_{0.52}As T1 ASQW structures to match ground state interband transition frequencies, resulting in total well widths of 10 nm, identical Al_{0.48}In_{0.52}As barriers, and similar well material compositions. Thus, direct comparisons between the T2 AQW and previously studied ACQW and T1 ASQW structures were possible. Comprehensive material parameters can be found in [supplementary material](#).

Figure 2 shows the $|\chi_{xxz}^{(2)}|$ and $|\chi_{zzx}^{(2)}|$ tensor elements as a function of T2 AQW well width asymmetry $l_1/(l_1 + l_2)$ for nearly resonant optical rectification (OR). The $\chi_{xxz}^{(2)}$ tensor is vanishingly small for nearly resonant OR and is not shown. Nearly resonant OR is a simple application of interband resonance^{18,24} to enhance $\chi^{(2)}$ where the input optical fields ω_1 and ω_2 are equal in magnitude and opposite in sign and are detuned slightly from the first interband resonance. Mathematically, nearly resonant OR is described by $\omega_1 = \omega_0 - \delta\omega$ and $\omega_2 = -\omega_1$, where ω_0 is the interband transition frequency and $\delta\omega$ is a detuning energy. Typical operation, therefore, utilizes small detuning $\delta\hbar\omega$ of ~ 10 – 100 meV so that absorption losses are minimal. However, the magnitude of $\chi^{(2)}$ is enhanced.^{2,18}

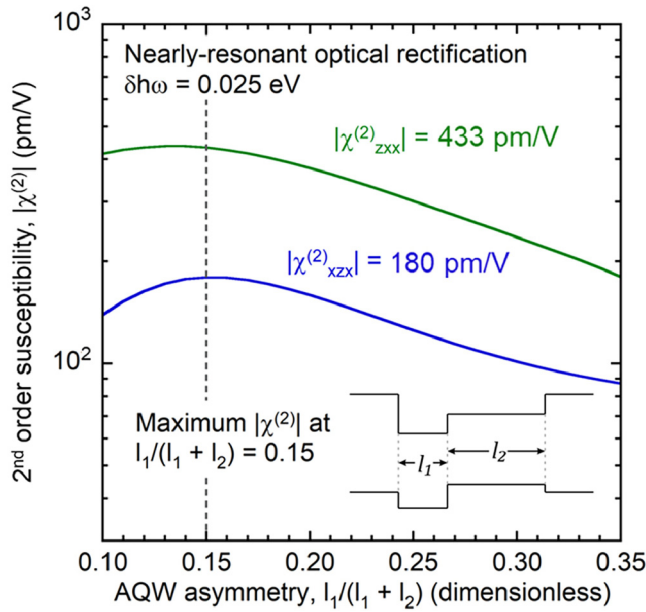


FIG. 2. Second-order susceptibility tensor elements $|\chi_{zxz}^{(2)}|$ (blue curve) and $|\chi_{zxx}^{(2)}|$ (green line) as a function of the well width asymmetry $l_1/(l_1 + l_2)$ in the InP/Al_{0.38}Ga_{0.10}In_{0.52}As T2 AQW structure. The total well width $l_1 + l_2$ is held constant at 10 nm. Nearly resonant optical rectification (OR) with $\omega_1 = \omega_0 - \delta\hbar\omega$ and $\omega_2 = -\omega_1$ is shown, where the transition frequency ω_1 is detuned by $\delta\hbar\omega = 25$ meV from the interband resonant transition frequency ω_0 . The $\chi_{zxz}^{(2)}$ tensor element reaches a maximum for $l_1 = 1.5$ nm.

The well widths of an ACQW and a T1 ASQW lattice-matched to InP are selected to yield the same interband transition energy $\hbar\omega_0 = 1.3$ eV as in the T2 AQW structure optimized in Fig. 2. The resulting quantum well potential profiles are illustrated

in Fig. 3, along with the electron, heavy hole, and light hole envelope wavefunctions squared.

The transition energies, envelope wavefunction overlap, interband transition dipole matrix element, and interband charge shift are shown in Table I for each interband transition in the three AQWs. Compared to the ACQW and ASQW, the T2 AQW exhibits moderate overlap between all quantum-confined states. Most importantly, the T2 AQW exhibits a large $\Delta z_{12,gm}$ between the e_1 -hh₁ and e_1 -lh₁ states. In contrast, $\Delta z_{12,gm}$ between the first quantum-confined states is minimal in the T1 ASQW and ACQW (also see Fig. 3).

Practically speaking, the application of second harmonic generation (SHG) makes little sense for mid-IR due to the limited efficiency and availability of powerful laser sources. Therefore, the interband $\chi^{(2)}$ tensor elements for the three AQW structures are only calculated for (a) nearly resonant OR, (b) singly resonant difference frequency generation (DFG) at 25 THz (0.103 eV), and (c) doubly resonant DFG near the e_2 - e_1 ISB transition. Singly resonant DFG makes use of the large increase in $\chi^{(2)}$ in interband type-II AQWs as the field frequency approaches the interband resonance and is suitable for the generation of THz radiation from a near-infrared pump laser.²¹ Doubly resonant DFG enhances $\chi^{(2)}$ further by placing the output field component ω_2 in resonance with an ISB transition.^{19,21}

Figure 4 compares the IB $\chi^{(2)}$ tensor elements of the three AQW structures. The T2 AQW structure (green bars) exhibits enhanced $\chi^{(2)}$ in all applications except for the small $\chi_{zxx}^{(2)}$ tensor element for singly resonant 25 THz DFG. In particular, the $\chi_{zxz}^{(2)}$ tensor element is enhanced by approximately 1 order of magnitude in the T2 AQW vs the ACQW and T1 ASQW structures. The ACQW and T1 ASQW structures exhibit approximately equal $\chi^{(2)}$ values, which can be intuitively understood from their nearly identical envelope wavefunction overlaps and dipole matrix elements shown in Table I. The enhancement of $\chi^{(2)}$ in the T2 AQW is a result of the large center of charge difference between the e_1 -hh₁ and e_1 -lh₁ states, in contrast to the ACQW and T1 ASQW

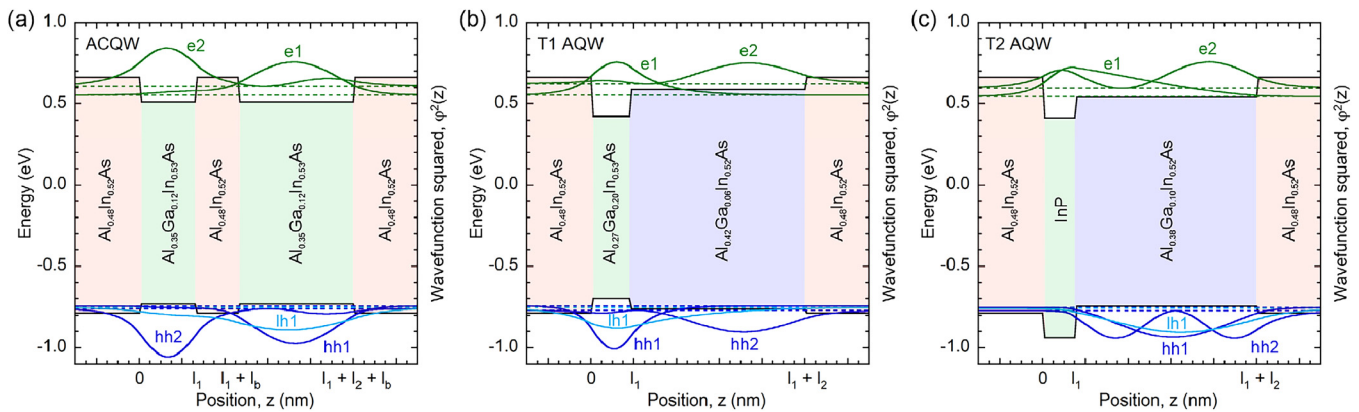


FIG. 3. Schematic layer structure of band offsets and envelope wavefunctions squared for (a) ACQW structure, (b) T1 ASQW structure, and (c) T2 AQW structure. All three quantum well structures are lattice-matched to InP with a ground state e_1 -hh₁ transition energy of 1.300 eV.

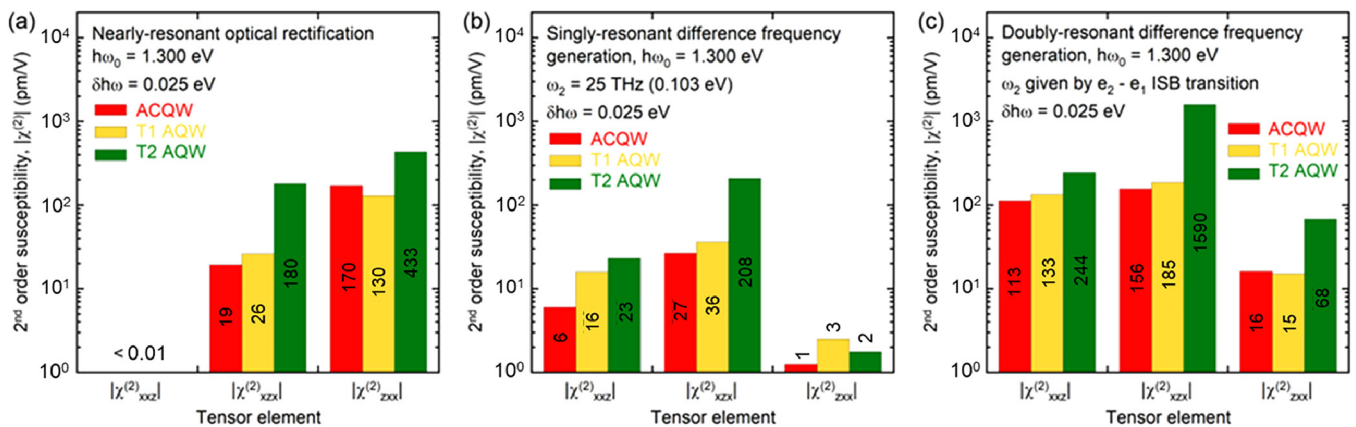
TABLE I. Interband transition energies, envelope wavefunction overlap, interband dipole matrix elements, and difference of center of charge for the AQW structures shown in Fig. 3.

Structure	Interband transition	Transition energy, $\hbar\omega_{12,gm}$ (eV)	Envelope wavefunction overlap, $\langle\psi_{b1,g} \psi_{b2,m}\rangle$	Interband dipole matrix element, $ \hat{x} \times \mathbf{r}_{12,gm} $ (Å)	Charge shift, $\Delta z_{12,gm}$ (nm)
ACQW [Fig. 3(a)]	e_1 - hh_1	1.300	0.993	5.01	-0.25
	e_1 - lh_1	1.307	0.971	2.88	0.61
	e_1 - hh_2	1.315	0.097	4.96	4.98
	e_2 - hh_1	1.354	-0.093	4.81	-4.83
	e_2 - lh_1	1.362	0.173	2.76	-3.97
	e_2 - hh_2	1.369	0.985	4.76	0.39
T1 ASQW [Fig. 3(b)]	e_1 - hh_1	1.298	0.984	4.98	0.27
	e_1 - lh_1	1.309	0.962	2.85	-0.49
	e_1 - hh_2	1.325	-0.136	4.88	-5.36
	e_2 - hh_1	1.367	0.093	4.73	5.80
	e_2 - lh_1	1.378	-0.243	2.71	5.04
	e_2 - hh_2	1.395	0.990	4.64	0.17
T2 AQW [Fig. 3(c)]	e_1 - hh_1	1.300	0.771	4.97	-3.94
	e_1 - lh_1	1.307	0.790	2.86	-4.53
	e_1 - hh_2	1.315	0.337	4.90	-4.50
	e_2 - hh_1	1.354	-0.617	4.78	-0.17
	e_2 - lh_1	1.362	-0.598	2.75	-0.76
	e_2 - hh_2	1.369	0.586	4.71	-0.73

structure. The electron and hole envelope wavefunction peaks in the T2 AQW are confined in separate well layers at each subband. The spatial separation of the electron and hole wavefunctions within each subband contributes to a large interband electric dipole moment in the type-II aligned structure and thus increases the achievable $\chi^{(2)}$. This spatial separation of wavefunction peaks within each subband is absent in the type-I aligned structures

where significant cancellations between inter- and intra-band transition dipole moments occur.²¹

The interband second-order susceptibility can be further enhanced by optimizing the asymmetric quantum well widths and band offsets. The conduction band offset ($\Delta E_{c2,c1}$) and valence band offset ($\Delta E_{v2,v1}$) are defined in Eqs. (6a) and (6b) such that $\Delta E_{c2,c1}$ and $\Delta E_{v2,v1}$ are defined as step well band edges subtracted

**FIG. 4.** Second-order susceptibility tensor elements $|\chi_{xxx}^{(2)}|$, $|\chi_{xzx}^{(2)}|$, and $|\chi_{zxx}^{(2)}|$ for (a) nearly resonant optical rectification, (b) singly resonant difference frequency generation at 25 THz (103 meV), and (c) doubly resonant difference frequency generation at the e_2 - e_1 ISB transition, all at a detuning energy $\delta\hbar\omega = 25$ meV.

by the InP band edges. There should exist a global maximum (and numerous local maxima) in the $\chi^{(2)}$ tensor elements as l_1 and $\Delta E_{c2,c1}$ and $\Delta E_{v2,v1}$ are varied. To maintain the type-II band alignment, the conduction band offsets $\Delta E_{c2,c1}$ vary from 0.025 to 0.250 eV ($\Delta E_{c2,c1} > 0$), while valence band offsets $\Delta E_{v2,v1}$ vary from 0.180 to 0.300 eV ($\Delta E_{v2,v1} > 0$), and the electron well width l_1 varies from 1 to 8 nm. The corresponding ground state e_1 -hh₁ transition energy E_0 as a function of electron well width l_1 , conduction band offset $\Delta E_{c2,c1}$, and valence band offset $\Delta E_{v2,v1}$ ranges from 1.180 to 1.377 eV, smaller than the bandgap of Al_{0.48}In_{0.52}As. This confirms that the e_1 -hh₁ interband transition is confined within the quantum well. The ground state transition energies vary by less than 0.2 eV, and therefore the influence of this variation on the magnitude of $\chi^{(2)}$ is minimal,

$$\Delta E_{c2,c1} = E_{c2} - E_{c1}, \quad (6a)$$

$$\Delta E_{v2,v1} = E_{v2} - E_{v1}. \quad (6b)$$

The second-order susceptibility tensor elements increase with narrower interband transition energies as seen in the denominator of Eq. (3). Therefore, a fair comparison of tensor elements between structures with varying interband transition energies can be achieved by normalizing the $\chi^{(2)}$ tensor values by the interband transition energies $E_0 = \hbar(\omega_{e1} - \omega_{hh1})$. The resulting figure of merit $|\chi^{(2)}| \times E_0$ has units of length $\times q$ (picometers $\times q$). With the parameters listed above, maximum normalized tensor elements $|\chi_{xxz}^{(2)}| \times E_0$ and $|\chi_{zzx}^{(2)}| \times E_0$ are calculated for nearly resonant OR (see Fig. 5). Compared with the original T2 AQW design demonstrated in Fig. 3(c), the normalized $\chi_{xxz}^{(2)}$ tensor is again vanishingly small, but the normalized $\chi_{xxz}^{(2)}$ and $\chi_{zzx}^{(2)}$ are approximately 2–3 times larger. This optimization study of T2 AQW structures suggests that narrower electron and wider hole wells give larger second-order susceptibility tensor elements than wider electron and narrower hole wells. Larger conduction band and valence band offsets tend to enhance the $\chi_{xxz}^{(2)}$ tensor element but reduce the $\chi_{zzx}^{(2)}$ tensor element. These general guidelines suggest that the optimum T2 AQW design will vary depending on the geometry of the incident and the extracted electric fields on a nonlinear optical device and thus the $\chi^{(2)}$ tensor element utilized.

The resonant enhancement achieved by the interband nonlinearity is smaller than that achieved for intersubband nonlinearity with $\chi_{zzz}^{(2)}$ up to $\sim 10^5$ pm/V demonstrated for second harmonic generation in type-I GaInAs/AlInAs ACQWs.¹⁴ However, the IB nonlinearity has the advantage of engendering large off-diagonal $\chi^{(2)}$ tensor elements, which enable a nonlinear optical response to both s- and p-polarized electric fields. In contrast, the $\chi_{zzz}^{(2)}$ tensor for the ISB nonlinearity is only excited by electric fields polarized normal to the growth plane (s-polarization).^{9,12,14,21} There is a notable similarity between THz/FIR generation in type-II quantum wells (QWs) and the shift current in materials without inversion centers,^{28–31} like ferroelectrics,³² involving electron excitation and real-space charge transfer. Specifically, exciting type-II QWs above the absorption edge causes charge movement by ΔZ_{11} and time-dependent polarization, generating shift current. However, shift current differs from second-order susceptibility in this study because it excites real carriers, constrained by recombination times.

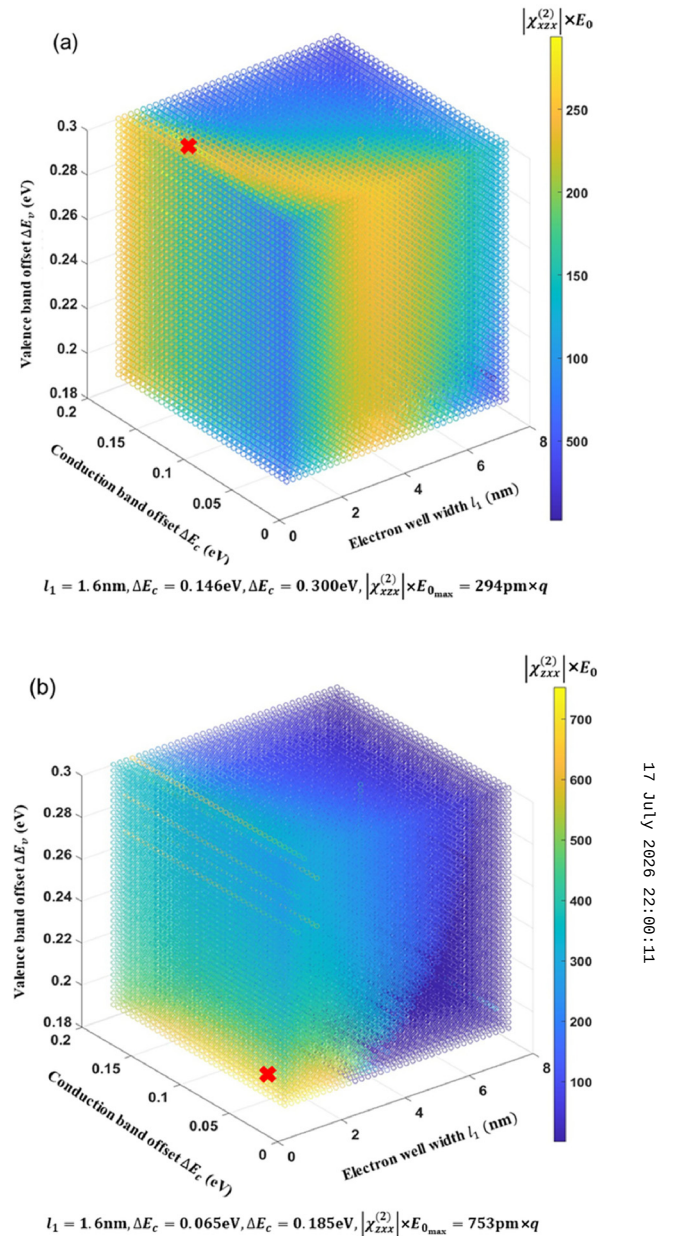


FIG. 5. Normalized second-order susceptibility tensor elements (a) $|\chi_{xxz}^{(2)}| \times E_0$ and (b) $|\chi_{zzx}^{(2)}| \times E_0$ for nearly resonant OR at constant detuning energy $\delta\hbar\omega = 25$ meV as a function of T2 AQW width l_1 and band offsets ΔE_c and ΔE_v . The optimized parameters for all three tensor elements are marked with red crosses which give the maximum normalized values.

Hence, although the leading edge of the shifty current pulse can be sharp, it is always followed by a long tail whose duration is recombination time which tends to be notably prolonged in type-II QWs due to the limited overlap of wavefunctions. For example, if one attempts to generate monochromatic THz/FIR radiation by beating

two near IR waves, the long lifetime of carriers will not allow one to reach even 10 GHz. Furthermore, photogenerated electrons and holes create a DC field that alters the band and wavefunctions, reducing and saturating the shift current. Our approach involves operation with a significant detuning (25 meV compared to a 10 meV broadening) from the absorption edge, thereby avoiding the generation of real carriers and instead utilizing “virtual” carriers. As a result, the operational speed is not restricted by relaxation times anymore and can reach 10s and 100s of THz. Moreover, without the presence of real carriers, no saturation in efficiency is anticipated, though, in less-than-ideal materials, some real carrier generation may occur, reducing efficiency.

The IB nonlinearity enables additional geometries for quasi-phase matching using domain inversion and additional operating wavelengths for photonic devices that are inaccessible by the extensively studied ISB nonlinearity.³³ Waveguides, normal incidence with gratings or metasurfaces can be further considered to achieve phase matching with normal incident light.^{34,35} Additionally, exploiting phase matching in the vicinity of phonon resonance is also a possible solution, capitalizing on anomalous dispersion.³⁶ It is important to note that such radiation can hardly be confined using dielectric materials. However, it can be confined using metal cladding layers as demonstrated by many groups. It is also noteworthy that by inverting the symmetry of the type-II quantum wells, it becomes possible to achieve matching between different order modes. In the context of generating higher frequencies, specifically near-infrared (IR) frequencies, conventional dielectric waveguides are typically employed. In the context of generating higher frequencies, specifically near-infrared (IR) frequencies, conventional waveguides are typically employed. There is not much difference between dielectrics and semiconductors when it comes to DFG. DFG has been successfully demonstrated in $\text{Al}_{0.5}\text{Ga}_{0.5}\text{As}$ waveguides confined by $\text{Al}_{0.6}\text{Ga}_{0.4}\text{As}$, where phase matching has been attained using periodic domain inversion—that type of phase matching is easily achievable for type-II ACQW by reversing the growth sequence.³⁷ Similarly, DFG was previously measured in GaAs/AlGaAs waveguides.³⁸ In our studies, the waveguide cladding can utilize a high Al content in the AlGaInAs quaternary alloy, which can also be lattice-matched to InP. There is also a possibility of using waveguide birefringence to achieve phase matching in GaAs waveguides.³⁹ Furthermore, the phase matching in semiconductor waveguides can extend even into the THz region as demonstrated before.⁴⁰ THz generation has been demonstrated in GaAs metasurface where the THz emission is normal to the surface, and phase matching is not required due to the thin thickness of the metasurface.⁴¹ Conclusively, we believe many strategies can be employed in order to achieve efficient DFG over a wide range of frequencies.

IV. CONCLUSIONS

Asymmetric quantum wells with type-II band alignment exhibit enhanced interband second-order susceptibility $\chi^{(2)}$ compared with conventional type-I aligned structures. Direct comparison between type-I and type-II band alignment is accomplished by designing structures with an identical total well width of 10 nm and interband transition energy of 1.3 eV using the lattice-matched

InP/AlInAs/AlGaInAs materials system. Further optimization of the type-II asymmetric quantum well width, conduction band offset, and valence band offset identifies global maxima in $\chi^{(2)}$ for nearly resonant optical rectification. The optimization study suggests that comparatively narrow electron well widths and large valence band offset between electron and hole wells are key to achieving high interband $\chi^{(2)}$. The $\chi^{(2)}$ enhancement results from the spatial separation between electron and hole wavefunction peaks into separate well layers in the type-II structure. The separation of electron and hole wavefunctions within each subband results in larger electric dipole moments than can be achieved in type-I aligned structures. Additionally, operation near the interband resonance at a suitable detuning energy of 25 meV greatly enhances $\chi^{(2)}$ by 1–2 orders of magnitude over natural bulk crystals such as LiNbO_3 , KTP, or GaAs. The InP/AlGaInAs type-II asymmetric quantum well is composed of mature III–V compound semiconductor materials, which can be grown by standard techniques such as molecular beam epitaxy or chemical vapor deposition. These properties recommend type-II asymmetric quantum wells for future ultra-compact on-chip integrated photonics.

SUPPLEMENTARY MATERIAL

Detailed derivation and calculation of dipole matrix elements, material parameters, calculations of absorption coefficients, and calculation of ground state transition energy of T2 AQW are included in the supplementary material.

ACKNOWLEDGMENTS

The authors acknowledge the funding support from Applied Materials and the National Science Foundation (Grant No. 1741694). The authors acknowledge Research Computing at Arizona State University for providing computing and storage resources that have contributed to the research results reported within this paper.

AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

S.T.S. and Z.J. contributed equally to this work and most of the writing of the manuscript. S.T.S., X.L., and X.Q. carried out the theoretical derivations, Z.J. did the computations, J.K. laid out the theory framework, and Y.H.Z. proposed the original idea to use T2 AQW and contributed to the editing of the manuscript.

Stephen T. Schaefer: Conceptualization (lead); Data curation (equal); Formal analysis (equal); Methodology (equal); Writing – original draft (lead); Writing – review & editing (equal). **Zheng Ju:** Data curation (equal); Formal analysis (equal); Investigation (equal); Methodology (equal); Software (equal); Writing – original draft (lead); Writing – review & editing (equal). **Xiaoyang Liu:** Data curation (equal); Methodology (equal); Software (equal). **Xin Qi:** Formal analysis (supporting); Investigation (supporting);

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Methodology (supporting); Writing – review & editing (equal). **Jacob Khurgin**: Conceptualization (equal); Formal analysis (equal); Supervision (equal); Writing – review & editing (equal). **Yong-Hang Zhang**: Formal analysis (equal); Funding acquisition (equal); Supervision (equal); Writing – review & editing (lead).

DATA AVAILABILITY

The data that support the findings of this study are available within the article and its [supplementary material](#).

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